

The scalability of IoT-enabled soil probes makes them suitable for both small-scale and large-scale agricultural operations, providing versatile solutions for modern farming.

Weed management with IoT

An automated weed removal robot powered by IoT is a state-of-the-art solution for precise and efficient weed control in agriculture. This robot seamlessly combines robotics, sensors, and connectivity to identify and eradicate weeds while significantly reducing the need for herbicides and manual labour.

IoT-based robots excel in delivering precise and targeted weed control, thereby minimising herbicide usage and mitigating potential harm to crops. These autonomous machines cover expansive areas swiftly, resulting in time and labour cost savings. The data gathered by the robot helps form weed management strategies.

The reduction in herbicide use and the enhancement of weed control practices foster environment-friendly and sustainable farming practices. Over time, the decreased reliance on herbicides and reduced labour costs leads to substantial savings for farmers. IoT-based weed removal robots are versatile and can be tailored to different field sizes and crop types.

Here's how an IoT-based weed removal robot operates.

Sensors and data collection:

Equipped with various sensors including cameras, infrared sensors, and GPS, the robot scans the field, discerning weeds from crops. Continuous data collection includes image capture and assessment of crop and weed health and growth stages. This data is then processed and analysed in real-time or transmitted to a central system or cloud platform via IoT connectivity.

Weed identification: Utilising a deep learning-based convolutional neural network (CNN) algorithm,

the robot accurately identifies and classifies weeds based on their physical characteristics such as appearance, size, colour, and growth patterns. The system boasts the ability to distinguish between crops and weeds with exceptional precision.

Precision herbicide application:

Once weeds are pinpointed, the robot employs precision application technology, directing herbicides solely towards the identified weeds. This precise targeting minimises herbicide usage and chemical exposure to crops. Some robots opt for mechanical methods like cutting, tilling, or uprooting to eradicate weeds without resorting to herbicides.

Autonomous navigation: Relying on GPS and mapping data, the robot autonomously navigates within the field. It adheres to predetermined paths or employs obstacle detection and avoidance mechanisms to manoeuvre around obstructions.

Remote monitoring and

control: Farmers and agronomists can remotely oversee the robot's progress, making real-time adjustments via a web-based or mobile application. This remote control allows for route modifications, herbicide application adjustments, or the pausing of operations when needed.

Data logging and analysis: The robot systematically records data pertaining to weed density, distribution, and treatment effectiveness. Historical data is securely stored, facilitating comprehensive analysis and enabling farmers to make informed decisions about weed management and future optimisation.

Weather integration: The robot may seamlessly integrate with weather forecasting services to make well-informed decisions regarding operation timing, accounting for weather conditions such as wind speed and rainfall forecasts.

IoT-based irrigation systems

Data is transmitted via IoT connectivity to a central system for analysis.

Based on this data, the system makes informed decisions about when and how much to irrigate a crop.

IoT-based irrigation systems offer water efficiency, energy savings, improved crop health, cost savings, environmental sustainability, data insights, and scalability. They prevent over-irrigation and water wastage, leading to higher crop yields. By conserving water and energy and reducing the risk of over-irrigation, these systems contribute to more sustainable and environment-friendly farming practices. IoT-based irrigation systems can be adapted for farms of various sizes, from small family farms to large commercial operations.

Farmers can remotely monitor and control the system through web-based or mobile applications, allowing real-time adjustments to irrigation schedules.

IoT services from Amazon

Amazon Web Services (AWS) offers a range of IoT services to help farmers and organisations build and manage IoT applications and solutions. These services ease the complexities of IoT, including device management, data collection and analysis, and connectivity.

AWS IoT Core: This is the central service for IoT on AWS. It provides the ability to connect IoT devices to the cloud securely and at scale. It supports MQTT and HTTP protocols for device communication and provides features for device authentication, authorisation, and message routing.

AWS IoT Device Management:

This service allows you to register, organise, monitor, and remotely manage IoT devices at scale. We can use it to maintain device information, push firmware updates, and track device health.